
Object-Centric Learning with Slot Attention

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Distributed Representations VS Object-Centric Representations

比如一个图像 encoder 输出一个 latent vector:

```
z = [0.12, -0.37, 0.81, ..., 0.05]
```

这里面通常没有一个维度可以简单解释成:

第 1 维 = 位置
第 2 维 = 颜色
第 3 维 = 形状

而是变成多个 slot:

```
image → [slot 1, slot 2, slot 3, slot 4]
```

理想情况下:

slot 1 → 人
slot 2 → 狗
slot 3 → 车
slot 4 → 背景

Distributed Representations

1. 分布式表示 (Distributed Representation)

把信息混合编码在一个向量的所有维度中，没有明确的“对象”边界

输入：一张包含多个物体的图像



Encoder

维度 1
维度 2
维度 3
维度 4
...
维度 D

分布式表示：一个高维向量 z

-0.21	0.73	-0.11	0.95	-0.37	...	0.42
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每个维度都同时包含了
汽车、狗、自行车、房子、背景等信息
很难解释某一维度具体表示什么

解码重建
Reconstruction



可视化：改变某一维度的值，整张图像会发生复杂变化（难以对应到具体物体）

维度 k 从 $-2 \rightarrow +2$



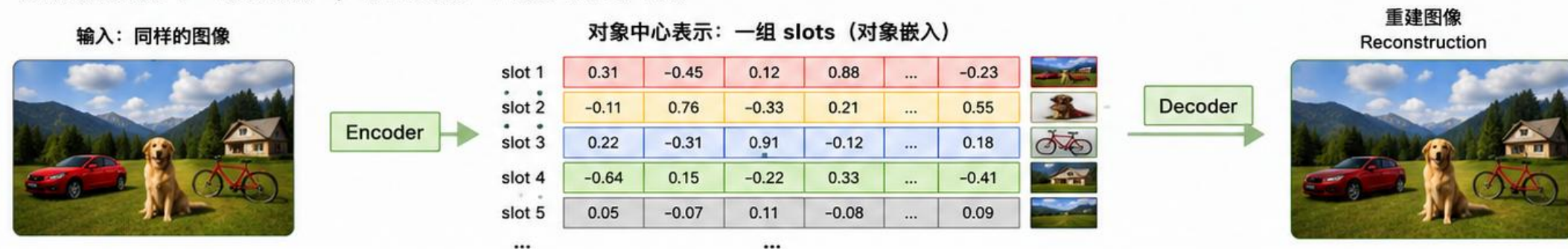
特点总结

- 没有明确的对象边界
- 所有信息混合在一起
- 表达能力强，但可解释性弱
- 不利于分析对象之间的关系

Object-Centric Representations

2. 对象中心表示 (Object-Centric Representation)

把输入拆解为多个“对象/成分”，每个对象用一个向量 (slot) 表示



每个 slot 主要关注一个对象 (可视化每个 slot 的注意力/掩码)



编辑单个 slot, 只改变对应的对象, 其他不变



特点总结

- 有明确的对象/成分边界
- 每个 slot 对应一个对象
- 可解释性强, 便于分析和干预
- 更适合建模对象关系与组合泛化

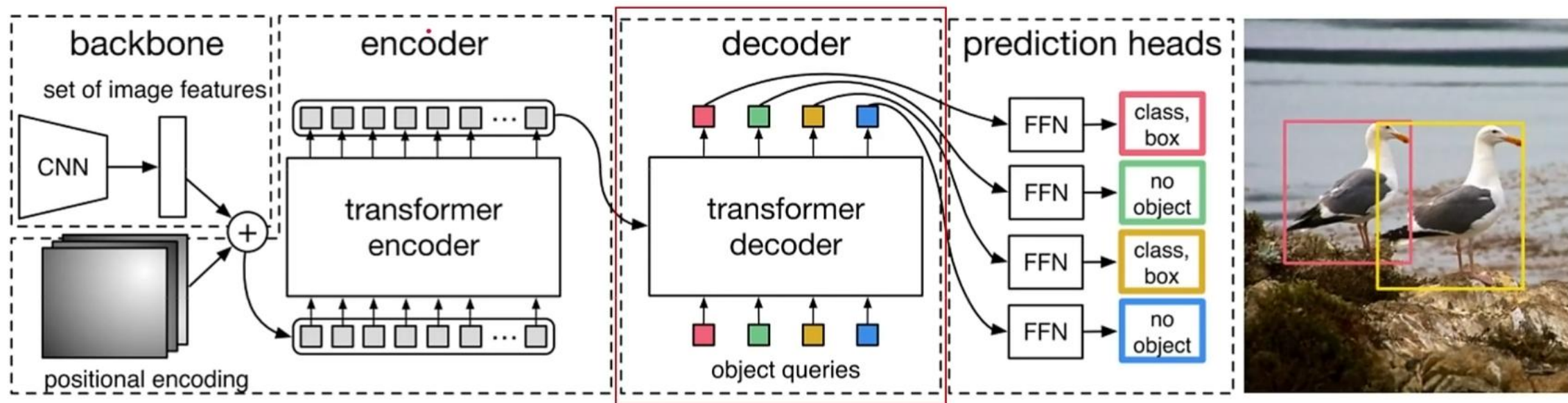
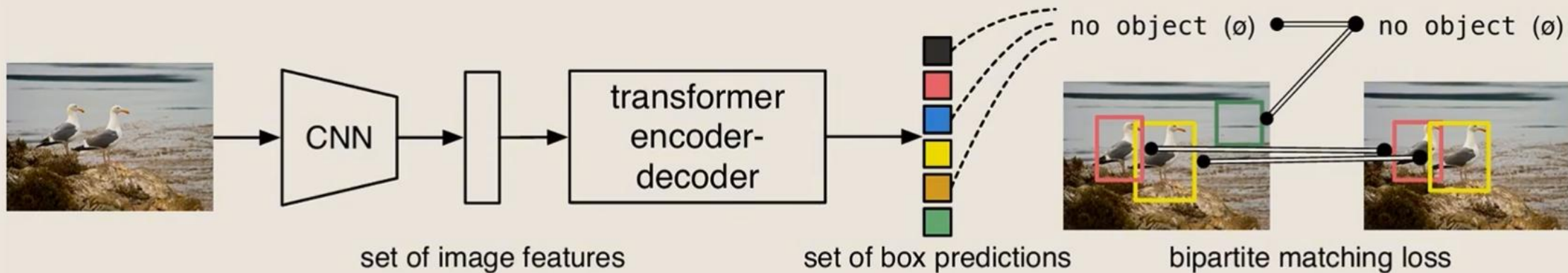
核心区别 一图总结

输入 → 一个向量 → 重建
信息混合在一起, 难以分解

VS.

对象中心表示:
输入 → 对象1 | 对象2 | ... | 对象N → 重建
信息被分解为多个对象/成分, 便于理解、分析和控制

DETR (DEtection TRansformer)

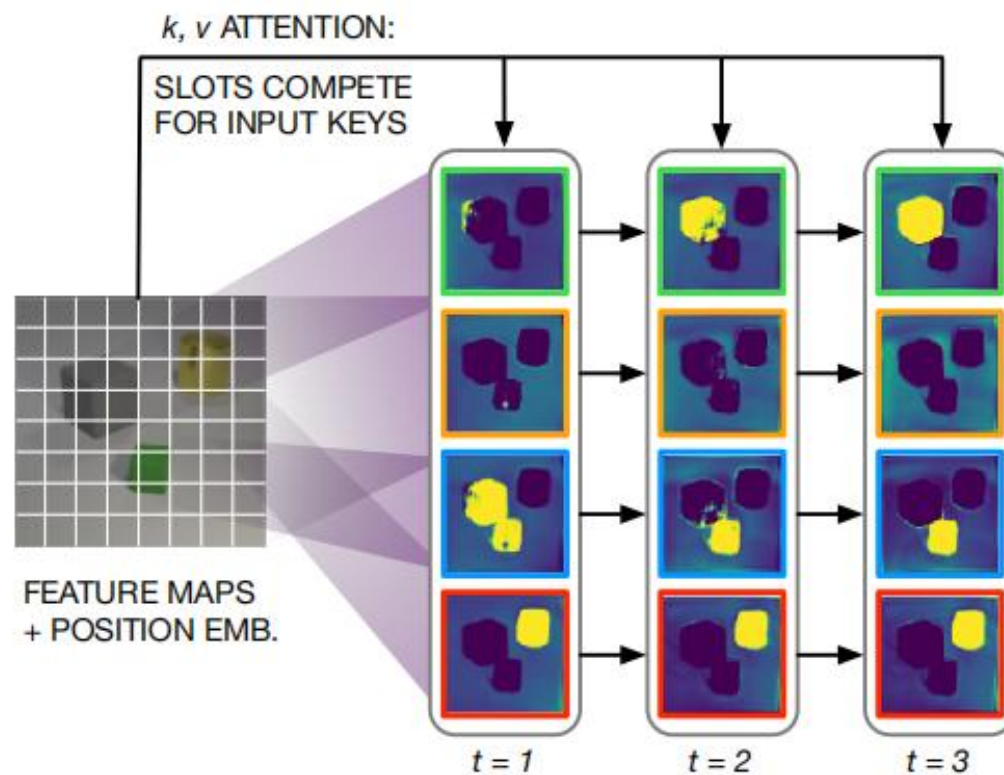


Slot Attention

Algorithm 1 Slot Attention module. The input is a set of N vectors of dimension D_{inputs} which is mapped to a set of K slots of dimension D_{slots} . We initialize the slots by sampling their initial values as independent samples from a Gaussian distribution with shared, learnable parameters $\mu \in \mathbb{R}^{D_{\text{slots}}}$ and $\sigma \in \mathbb{R}^{D_{\text{slots}}}$. In our experiments we set the number of iterations to $T = 3$.

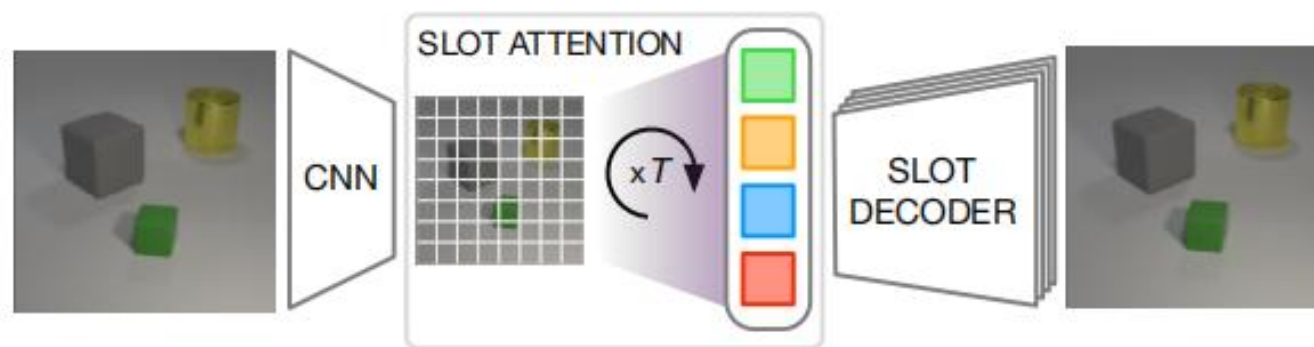
```
1: Input:  $\text{inputs} \in \mathbb{R}^{N \times D_{\text{inputs}}}$ ,  $\text{slots} \sim \mathcal{N}(\mu, \text{diag}(\sigma)) \in \mathbb{R}^{K \times D_{\text{slots}}}$ 
2: Layer params:  $k, q, v$ : linear projections for attention; GRU; MLP; LayerNorm (x3)
3:    $\text{inputs} = \text{LayerNorm}(\text{inputs})$ 
4:   for  $t = 0 \dots T$ 
5:      $\text{slots\_prev} = \text{slots}$ 
6:      $\text{slots} = \text{LayerNorm}(\text{slots})$ 
7:      $\text{attn} = \text{Softmax}(\frac{1}{\sqrt{D}} k(\text{inputs}) \cdot q(\text{slots})^T, \text{axis}='slots')$       # norm. over slots
8:      $\text{updates} = \text{WeightedMean}(\text{weights}=\text{attn} + \epsilon, \text{values}=v(\text{inputs}))$       # aggregate
9:      $\text{slots} = \text{GRU}(\text{state}=\text{slots\_prev}, \text{inputs}=\text{updates})$       # GRU update (per slot)
10:     $\text{slots} += \text{MLP}(\text{LayerNorm}(\text{slots}))$       # optional residual MLP (per slot)
11:  return  $\text{slots}$ 
```

Slot Attention module

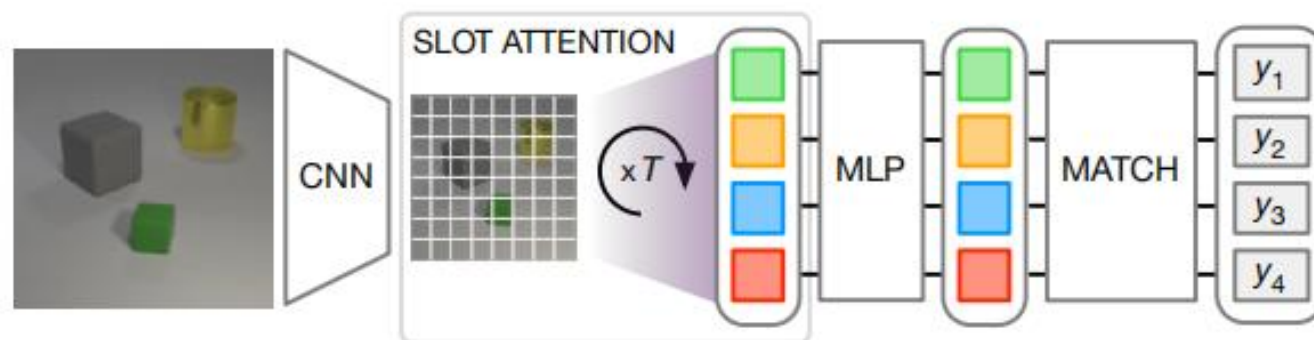


(a) Slot Attention module.

application 有监督 与 无监督



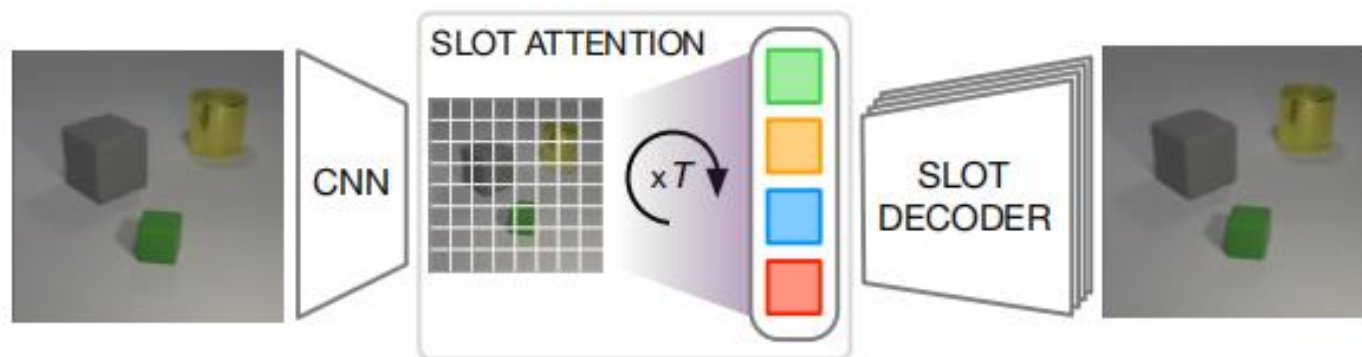
(b) Object discovery architecture.



(c) Set prediction architecture.

Object discovery architecture

image
↓
CNN encoder
↓
Slot Attention
↓
K 个 slots
↓
每个 slot 单独经过 decoder
↓
每个 slot 输出 RGB reconstruction + alpha mask
↓
所有 slots 合成最终重建图像



Set prediction architecture

image



CNN encoder



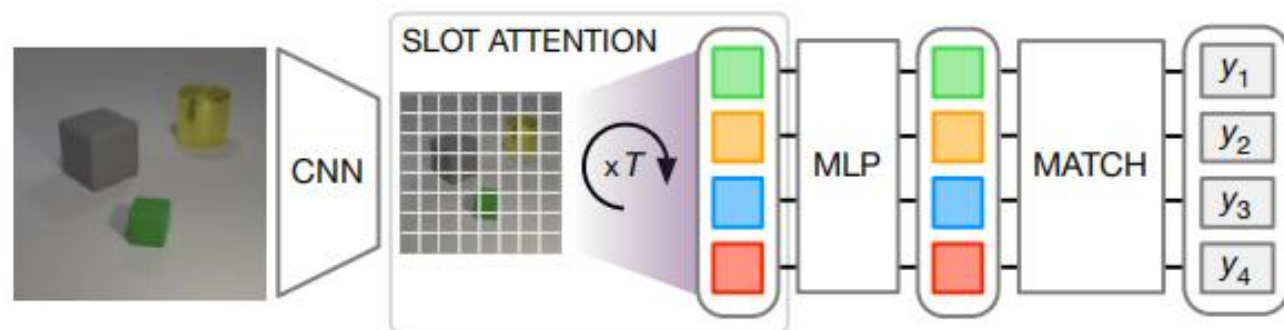
slot Attention



每个 slot 经过共享 MLP



预测一个物体的属性



(c) Set prediction architecture.

Results

	CLEVR6	Multi-dSprites	Tetrominoes
Slot Attention	98.8 ± 0.3	91.3 ± 0.3	$99.5 \pm 0.2^*$
IODINE [16]	98.8 ± 0.0	76.7 ± 5.6	99.2 ± 0.4
MONet [17]	96.2 ± 0.6	90.4 ± 0.8	—
Slot MLP	60.4 ± 6.6	60.3 ± 1.8	25.1 ± 34.3

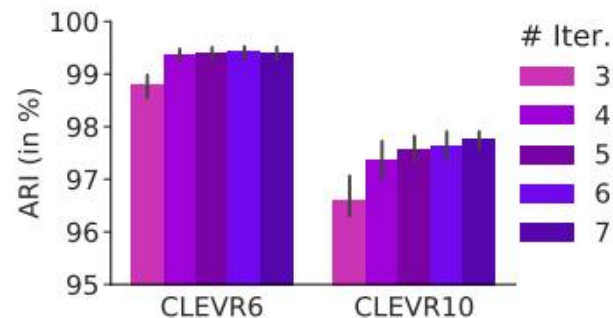
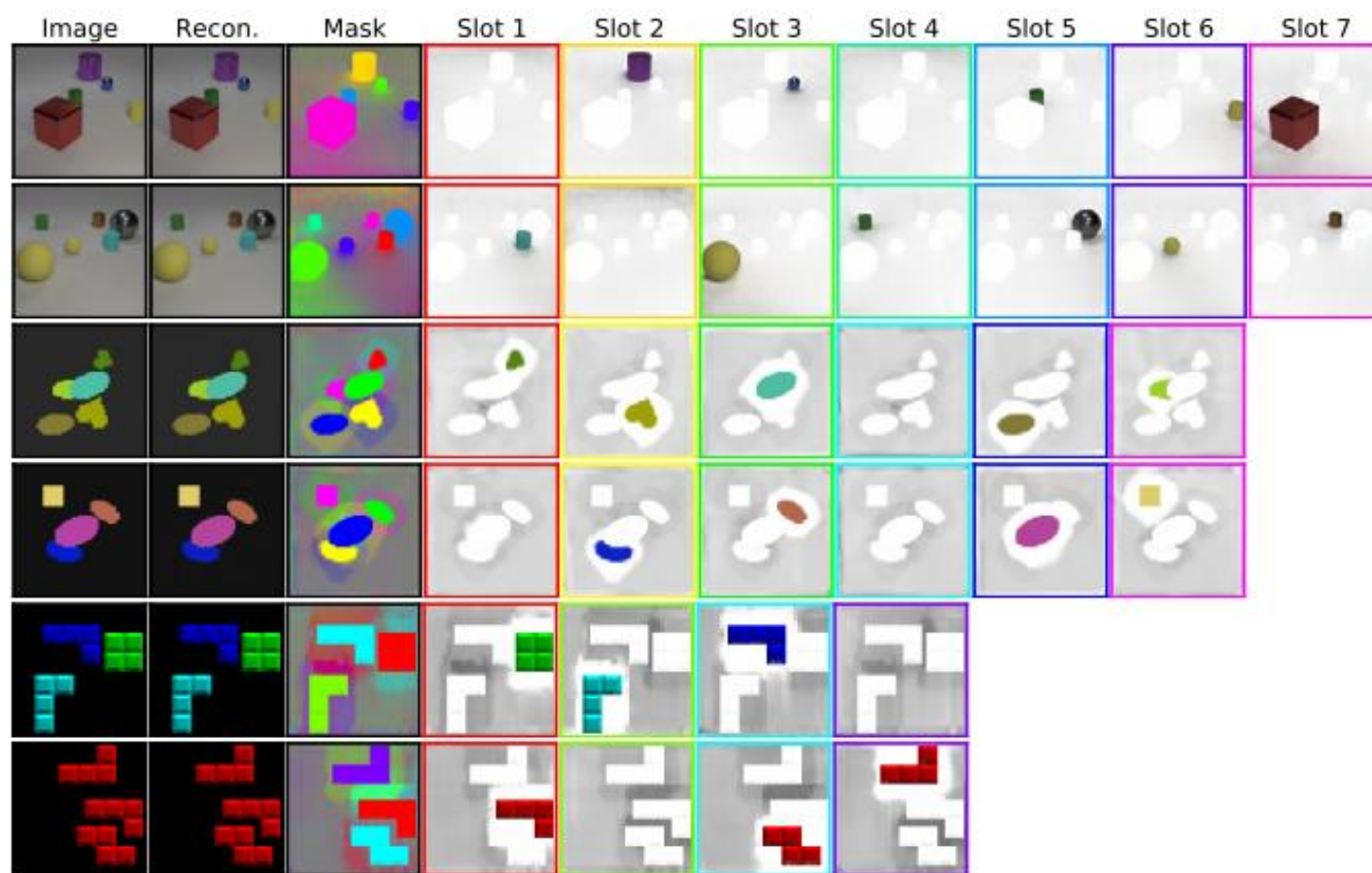
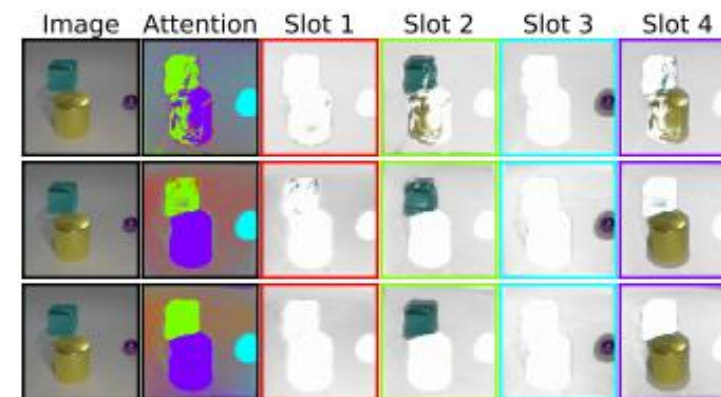


Table 1 & Figure 2: **(Left)** Adjusted Rand Index (ARI) scores (in %, mean $\pm$ stddev for 5 seeds) for unsupervised object discovery in multi-object datasets. In line with previous works [16, 17, 27], we exclude background labels in ARI evaluation. *denotes that one outlier was excluded from evaluation. **(Right)** Effect of increasing the number of Slot Attention iterations T at test time (for a model trained on CLEVR6 with $T = 3$ and $K = 7$ slots), tested on CLEVR6 ($K = 7$) and CLEVR10 ($K = 11$).

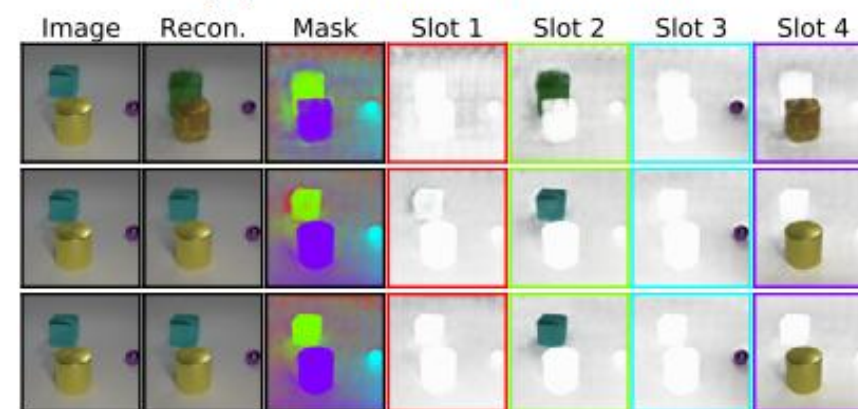
Results



(a) Decomposition across datasets.



(b) Attention iterations.



(c) Reconstructions per iteration.

Results

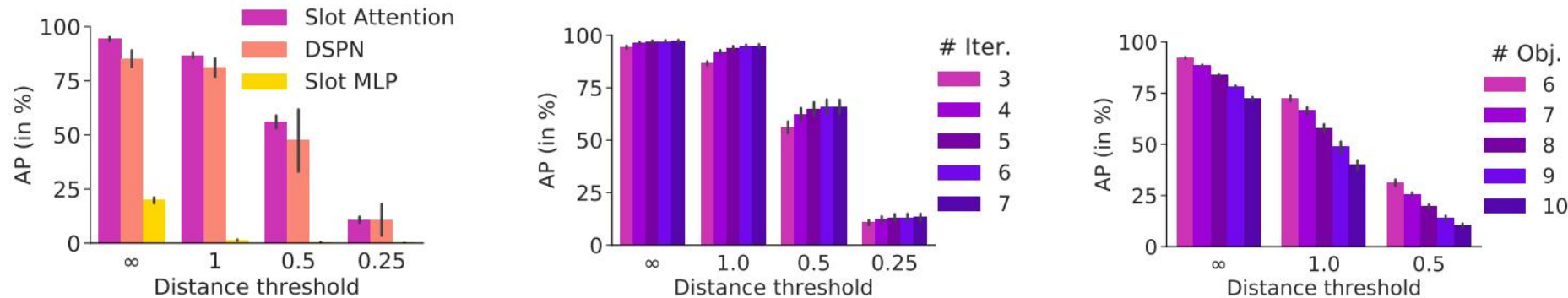


Figure 5: **(Left)** AP at different distance thresholds on CLEVR10 (with $K = 10$). **(Center)** AP for the Slot Attention model with different number of iterations. The models are trained with 3 iterations and tested with iterations ranging from 3 to 7. **(Right)** AP for Slot Attention trained on CLEVR6 ($K = 6$) and tested on scenes containing exactly N objects (with $N = K$ from 6 to 10).

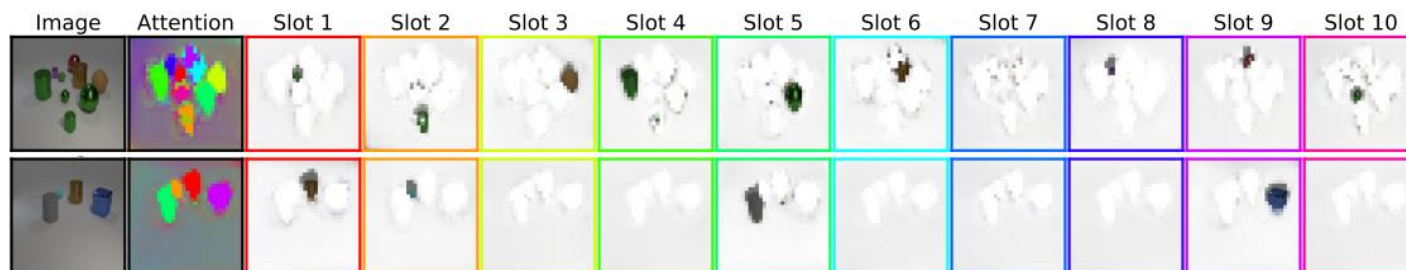


Figure 6: Visualization of the attention masks on CLEVR10 for two examples with 9 and 4 objects, respectively, for a model trained on the property prediction task. The masks are upsampled to 128×128 for this visualization to match the resolution of input image.

局限性

1.Slot Attention 本身并不知道什么是“物体”

2.背景没有特殊处理

3.对真实复杂图像仍有限制